AI Meets Arts

Bavarian-Georgian Summer School 2026
Hackathon Task

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Welcome!

We are happy to welcome you at Summer School 2026 !

Our main focus of the event is **"AI Meets Arts"** and especially **Deep Computer Vision**

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CAUCASUS UNIVERSITY

DAAD
Deutscher Akademischer Austauschdienst
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Learning takes time

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We have covered a lot of unfamiliar material in a very short time!

The pace has been demanding, and the learning path has not always been smooth. If some ideas still feel unclear or overwhelming, that is understandable.

- You are not expected to master every concept immediately and deeply
- Understanding grows through examples, questions, experiments, and revisiting ideas
- For your project, start with one clear question and build your understanding step by step

Thank you for your effort, curiosity, and willingness to keep learning!

Now it is time to...

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AI Meets Arts: your hackathon challenge ADAPT

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Develop and investigate an AI solution for an arts-related problem, using data acquired by your team: ADAPT

1. **A**cquire the data in the halls of CU
2. **D**efine a task and label the data accordingly
3. **A**nalyze, read, and understand relevant research papers and methods
4. **P**ut into practice the methods for your task and evaluate the results
5. **T**ell the story about your solution and findings

Final presentations: 15 October 2026, Amberg

Your project should connect an artistic question with a clearly defined AI task. Explain what you tried, what you learned, and what remains unresolved.

1. Acquire your own project data

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Each team must acquire the data for its project

- You make photographs of the art works at the University as well as collect the details, descriptions, or relevant metadata
- Collect the material and record its sources and any usage permissions
- Organize files using consistent IDs and a simple metadata table
- Inspect image quality, duplicates, diversity, and gaps in the collection

Questions to answer

- What does the collection represent, and what does it leave out?
- Is the material suitable for the task you want to investigate?

Stage output: an organized dataset and a short description of how it was acquired

2. Define the task and label the data

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Turn an artistic question into a concrete prediction or retrieval task.

Possible task	Desired output	Labels or reference information
Classification	A category for each image	Category labels
Image/text retrieval	A ranked list of relevant images	Relevant image–query matches
Object localization	Locations of selected motifs	Bounding boxes
Region segmentation	Spatial regions in an artwork	Masks or region annotations

- State the inputs, outputs, and intended use of your solution
- Write annotation rules and resolve ambiguous cases within the team
- Choose evaluation criteria and reserve examples for testing

Stage output: a task definition, annotation guide, labelled data, and evaluation plan

3. Analyze, read, and understand relevant research

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Choose methods because they fit your task and data

1. Search for relevant research papers and official implementations
2. Identify the central idea, required inputs, and model outputs
3. Determine what needs training and what supervision the method requires
4. Compare its assumptions with your dataset, computing resources, and time

Be prepared to explain

- Why is the method relevant? What makes it suitable for your problem?
- Which parts can you reuse, and which parts need adaptation?
- What are its limitations and likely failure cases?

Stage output: an annotated reading list and a justified method selection.

Cite the papers, pretrained models, and code you use.

Computational Resources

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- All lectures and the starter codes are in the Moodle course
  https://moodle.oth-aw.de/course/view.php?id=4657
- For simple experiments please use Google Colab
  https://colab.research.google.com/
- For more demanding experiments:
  - Every Team has access to https://ki-server.oth-aw.de/
  - The access is granted to OTH-AW member (through the university gitlab)
  - We have an Ollama Server with some free Vision-Language running on it
  - To view the models you can run in terminal:
    `OLLAMA_HOST=http://10.10.134.3:11434 ollama list`
- You are of course allowed to use Chatbots (but you are still responsible for your whole work!!!)

4. Put the methods to work and investigate the results

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Build a working pipeline and test your choices

1. Implement a simple baseline appropriate to your task
2. Adapt the selected method: preprocessing, prompts, features, prediction head, post-processing, or fine-tuning where justified
3. Compare the adapted solution with the baseline using consistent evaluation
4. Inspect successful cases, failures, and sensitivity to important design choices

Keep a record of your experiments: settings, model versions, results, and the reasons for changes

Stage output: a reproducible pipeline, baseline comparison, and analysis of findings

Keep test examples separate from model selection and tuning. With limited data, report uncertainty and individual errors. Training from scratch is not required.

5. Tell us: 15 October 2026 in Amberg

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Your presentation (20 minutes + 5 minutes questions) should cover:

1. **Question and task:** what did you investigate, and why does it matter?
2. **Data:** how did you acquire, organize, and annotate it?
3. **Research:** which papers and methods informed your approach?
4. **Adaptation:** what did you implement or change, and why?
5. **Evidence:** baseline comparison, results, examples, and failure cases.
6. **Findings:** what did you learn, what are the limitations, and what comes next?

Show your solution through a demonstration or clear input–output examples.
A well-explained negative result is also a finding.
Support your conclusions with evidence and distinguish your contribution from reused work.

Good Luck...

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Questions

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Thank you for your attention! I'm happy to answer any questions.

Contact

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Thank you!

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